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L04 Chihuahua or Muffin: Reflective Journal

This lab provided a practical introduction to how neural networks are trained for image classification. Unlike the traditional computer vision approaches studied in Module 03, where humans manually design feature extractors such as HOG or LBP, the neural network learned features directly from the image data. This reinforced one of the main ideas presented in the module: modern neural networks learn both the feature extraction process and the final classification task in an end-to-end manner rather than relying on handcrafted features.

One challenge I encountered during this assignment was distinguishing between code and explanatory text. In a previous activity, I accidentally included code as written discussion, which made me realize that understanding the purpose of each code block is just as important as executing it successfully. During this lab, I paid closer attention to how each component of the neural network contributed to the training process, including data transformations, dataloaders, the loss function, and the optimizer.

The training results showed gradual improvement across the three epochs. Training accuracy increased from 60% to approximately 73%, while validation accuracy improved from 73% to 80%. Although these results demonstrate that the network successfully learned patterns from the data, I was somewhat surprised by how uncertain many of its predictions remained. When examining the validation images, numerous classifications were close to 50% for both classes, indicating that the model often lacked confidence in its decisions. For that reason, I would describe the model's performance as acceptable but not particularly strong.

This observation helped me better understand one of the limitations of basic neural network architectures for computer vision tasks. While the network was able to learn directly from raw pixels, it still struggled to produce highly confident predictions on many images. This suggests that additional training, architectural changes, data augmentation, or more specialized image-processing networks could further improve performance. The exercise also demonstrated the importance of hyperparameters such as learning rate, number of epochs, and network structure, all of which influence the final results.

Image classification techniques similar to those explored in this lab have many practical applications. Examples include medical image analysis, facial recognition, manufacturing quality inspection, wildlife monitoring, and autonomous vehicle perception systems. Although the model developed in this lab was relatively simple, it demonstrated the same fundamental principles used in many real-world computer vision applications. The experience helped me better understand how image data can be transformed into meaningful predictions and why model reliability is important in practical settings.

Overall, this lab helped me connect the theoretical concepts from the module with a real implementation in PyTorch. Concepts such as the forward pass, loss calculation, gradient descent, and backpropagation became much clearer after observing how they worked together during training. The activity also highlighted the major shift from traditional computer vision methods to modern deep learning approaches, where feature learning and classification are performed simultaneously by the network itself.

References

McManus, P. (2026). *Neural Networks for Computer Vision: When Machines Stopped Following Our Recipes and Started Writing Their Own*. Houston City College, ITAI 1378 Module 04.

McManus, P. (2026). *Neural Networks for Computer Vision (Module 04 Slides)*. Houston City College.